

Introduction to UNIX and RNA-seq Repositories

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**Canadian Centre for
Computational
Genomics**



Outline

- Setup & Troubleshooting
 - UNIX availability check
 - Fix PATH / permissions
- Module 1 — UNIX Basics
 - File management and navigation
 - File inspection
 - Pipes and redirects
 - Text search
- Module 2 — GEO Basics
 - Accession types
 - Common files
 - FTP/HTTPS & Data Download

Setup & Troubleshooting

pwd

mkdir -p ~/workshop/data && cd ~/workshop

which bash zsh curl wget gzip tar || true

```
ztava@Zahra:~/workshop$ cd ~
ztava@Zahra:~$ pwd
/home/ztava
ztava@Zahra:~$ mkdir -p ~/workshop/data && cd ~/workshop
ztava@Zahra:~/workshop$ which bash zsh curl wget gzip tar || true
/usr/bin/bash
/usr/bin/curl
/usr/bin/wget
/usr/bin/gzip
/usr/bin/tar
```

wsl --install -d Ubuntu

wsl -l -v

wsl -d Ubuntu

exit or Ctrl+D → To exit

wsl --unregister "Ubuntu"

sudo apt update -y

sudo apt install -y curl wget gzip tar grep gawk sed

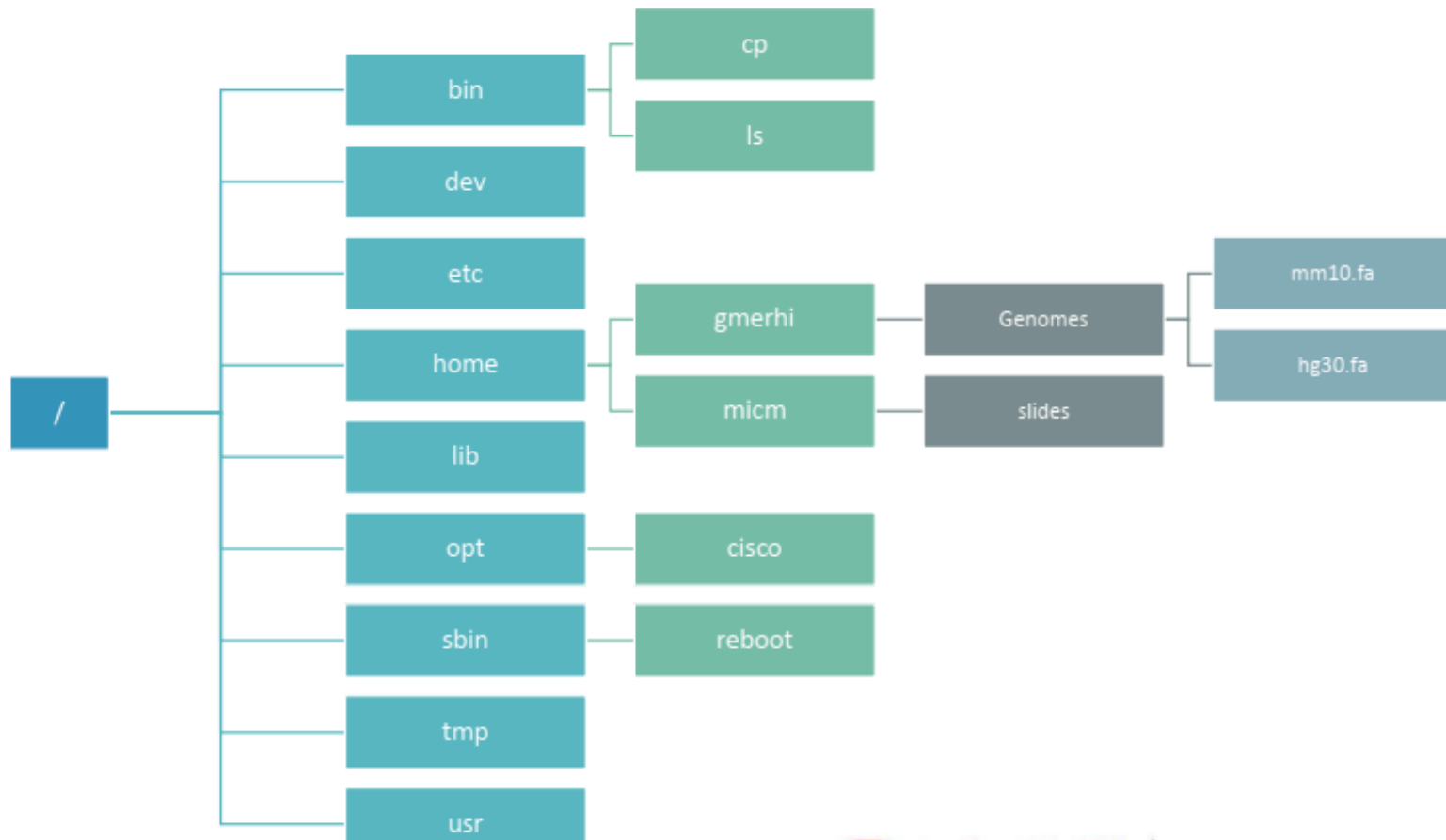
```
PS C:\Users\ztava> wsl --install -d Ubuntu
Downloading: Ubuntu
Installing: Ubuntu
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu'
Launching Ubuntu...
Provisioning the new WSL instance Ubuntu
This might take a while...
Create a default Unix user account: ztava
New password:
Retype new password:
No password has been supplied.
New password:
Retype new password:
passwd: password updated successfully
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

```
ztava@Zahra:/mnt/c/Users/ztava$ |
```

Module 1

UNIX Basics

File system structure



Keyboard shortcuts

Ctrl+A	Ctrl+E	Ctrl+C	q
Go to the start of the line	Go to the end of the line	Stop the current process	Exist a child process (less, more, etc)

Unix commands

- Programs built in the shell that perform specific actions



Basic commands

top

See active
processes and
the resources
they're using

% htop

man

Shows the
manual page of
a command

% man ls
% man cd
% man htop

history

List your
previous
commands

% history

clear

Clear your
terminal
window

% clear

File management & navigation

Command	What it does
<code>pwd</code>	print working directory — "where am I?"
<code>ls</code>	list files; <code>ls -la</code> shows hidden + details
<code>cd</code>	change directory; <code>cd ~</code> goes home, <code>cd ..</code> goes up
<code>mkdir</code>	make a new directory; <code>-p</code> makes parents as needed
<code>mv</code>	move or rename
<code>rm</code>	remove (no undo — be careful)

Special characters

.

..

*

~

Current
directory

Parent
directory

Match any and
all characters

Home
directory

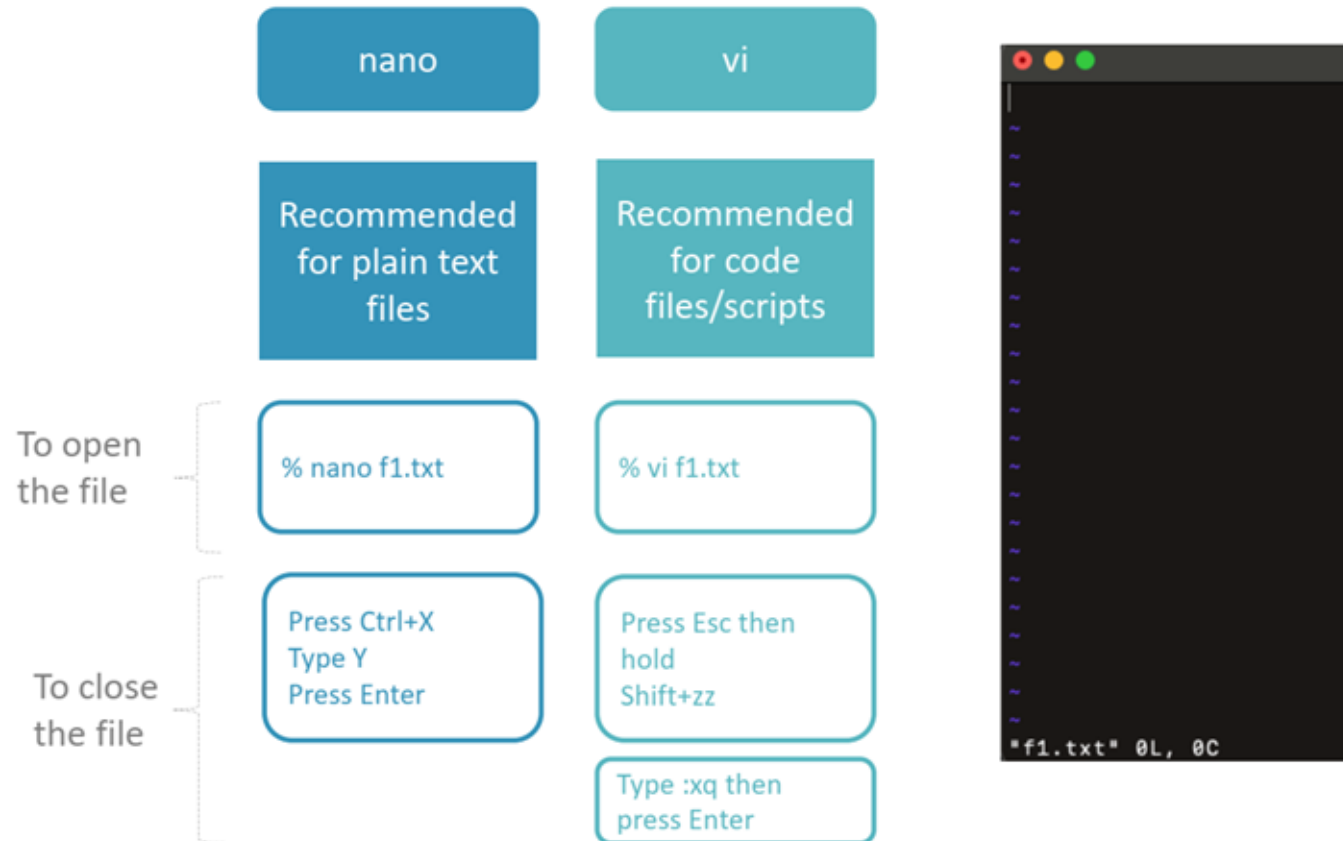
% ls .

% ls ..

% ls *

% ls ~

How to edit files in the terminal



File inspection

- `head -n 10 file` — first 10 lines
- `tail -n 5 file` — last 5 lines
- `less file` — scroll through; `q` to quit
- `wc -l file` — count lines

Pipes and redirects

- `|` — send the output of one command into the next
- `>` — write to a file (overwrite)
- `>>` — append to a file
- Example: `ls -la | wc -l` counts how many entries are in a folder

Exercise 1

You will:

1. Find out where you are and what's around you
2. Build a small project folder tree
3. Create files, write numbers into one, rename another
4. Combine files with `cat` and a pipe

Answer key: `Exercises/answers/Exercise1_unix.sh`

Pattern matching

grep

Search a
pattern in a file

```
% grep Male  
happiness.csv
```

Country Gender Mean N=

AT **Male** 7.3 471

AT Female 7.3 570

AT Both 7.3 1041

BE **Male** 7.8 468

BE Female 7.8 542

BE Both 7.8 1010

BG **Male** 5.8 416

BG Female 5.8 555

BG Both 5.8 971

Regular expressions

Groups and ranges

.	Any character except new line (\n)
(a b)	a or b
(...)	Group
(?:...)	Passive (non-capturing) group
[abc]	Range (a or b or c)
[^abc]	Not (a or b or c)
[a-q]	Lower case letter from a to q
[A-Q]	Upper case letter from A to Q
[0-7]	Digit from 0 to 7
\x	Group/subpattern number "x"

Character classes

\s	White space
\S	Not white space
\d	Digit
\D	Not digit
\w	Word
\W	Not word

Regular expressions

Anchors

^	Start of string, or start of line in multi-line pattern
\A	Start of string
\$	End of string, or end of line in multi-line pattern
\Z	End of string
\b	Word boundary
\B	Not word boundary
\<	Start of word
\>	End of word

Quantifiers

*	0 or more	{3}	Exactly 3
+	1 or more	{3,}	3 or more
?	0 or 1	{3,5}	3, 4 or 5

Special characters

^	[	.	\$
{	*	(	\
+	)	 	?
<	>		
The escape character is usually \			

Example

- All lines that start with a letter

```
% grep -E "^[A-Z]+" happiness.csv
```

- Lines that do not start with a letter

```
% grep -v -E "^[A-Z]+" happiness.csv  
% grep -E "^\W" happiness.csv
```

- Lines with a country from the list

```
% grep -f countries.txt happiness.complete.tsv
```

Exercise 3.1

1. Preview & row count (*zcat/gunzip*)

- Show the first 10 and last 5 lines.
- Report the number of data rows (exclude header).

2. Print the header with column indices so you can choose sample columns.

```
zcat GSE251845_htseq_raw_counts.csv.gz | head -n 1 | awk -F',' '{for(i=1;i<=NF;i++) printf("%d\t%s\n", i, $i)}'
```

```
ztava@Zahra:~/workshop$ zcat GSE251845_htseq_raw_counts.csv.gz | head -n 1 | awk -F',' '{for(i=1;i<=NF;i++) printf("%d\t%s\n", i, $i)}'
```

1	""
2	"24C_htseq.out"
3	"24N_htseq.out"
4	"27C_htseq.out"

3. Pick & extract

- Choose two sample columns (your choice) plus the gene ID column.
- Save to **subset_cols.csv**.

4. Sort (*sort -t',' -k3,3nr*)

- Sort **subset_cols.csv** descending by one of your sample columns.
- Keep the header; save as **sorted_subset.csv**.

Exercise 3.1

5. Low-information checks

- ANY-zero genes: genes with at least one zero across samples.

```
awk -F',' 'NR>1 && ($2==0 || $3==0){c++} END{print c}' # ANY-zero
```

- ALL-zero genes: genes with all zeros across samples.
- Also report highly expressed genes for one chosen sample column (e.g., counts ≥ 1000).

Exercise 3.2

Move/copy your downloaded `*.fastq` files into your working directory.

Count reads (headers) — no decompression needed (hint: @)

Find how many reads contain ambiguous base 'N'

Top 10 most common 6-mers at the 5' end

Motif search (choose a short motif, e.g., ACGTAC)

Take the first 100 reads

Module 1

GEO Basics

What is GEO?

- **Gene Expression Omnibus** — NCBI's public repository for functional genomics data
- Anything published with sequencing data ends up here (by NIH mandate)
- Two layers of data:
 - **Processed**: count matrices, normalized expression — ready to analyze
 - **Raw**: the original FASTQ reads — sitting in SRA, linked from GEO

GEO (Gene Expression Omnibus)

 [Resources](#)  [How To](#) 

[GEO Home](#) [Documentation](#) [Query & Browse](#) [Email GEO](#)

 **Notice**

Because of a lapse in government funding, the information on this website may not be up to date, transactions submitted via the website may not be processed, and the agency may not be able to respond to inquiries until appropriations are enacted. The NIH Clinical Center (the research hospital of NIH) is open. For more details about its operating status, please visit cc.nih.gov. Updates regarding government operating status and resumption of normal operations can be found at opm.gov.

Gene Expression Omnibus

GEO is a public functional genomics data repository supporting MIAME-compliant data submissions. Array- and sequence-based data are accepted. Tools are provided to help users query and download experiments and curated gene expression profiles.


Gene Expression Omnibus

Getting Started

- [Overview](#)
- [FAQ](#)
- [About GEO DataSets](#)
- [About GEO Profiles](#)
- [About GEO2R Analysis](#)
- [How to Construct a Query](#)
- [How to Download Data](#)

Tools

- [Search for Studies at GEO DataSets](#)
- [Search for Gene Expression at GEO Profiles](#)
- [Search GEO Documentation](#)
- [Analyze a Study with GEO2R](#)
- [Studies with Genome Data Viewer Tracks](#)
- [Programmatic Access](#)
- [FTP Site](#)
- [ENCODE Data Listings and Tracks](#)

Browse Content

Repository Browser

DataSets:	4348
Series: 	264424
Platforms:	27739
Samples:	8058260

Accession types — learn to read them

Prefix	Meaning	Example
GSE	Series — a whole study	GSE251845
GSM	Sample — one library in a study	GSM7980123
GPL	Platform — the sequencer / array model	GPL24676
SRR	Run — the actual raw read file in SRA	SRR390728
SRX / SRP	Experiment / Project — SRA-side groupings	—

Mnemonic: **GSE** → **GSM** → **SRR**. Study → Sample → Reads.

Status	Public on Apr 11, 2024
Title	Identification of unique gene expression and splicing events in early-onset colorectal cancer
Organism	Homo sapiens
Experiment type	Expression profiling by high throughput sequencing
Summary	Background: The incidence of colorectal cancer (CRC) has been steadily increasing in younger individuals over the past several decades for reasons that are incompletely defined. Identifying differences in gene expression profiles, or transcriptomes, in early-onset colorectal cancer (EOCRC, < 50 years old) patients versus later-onset colorectal cancer (LOCRC, > 50 years old) patients is one approach to understanding molecular and genetic features that distinguish EOCRC. Methods: We performed RNA-sequencing (RNA-seq) to characterize the transcriptomes of patient-matched tumors and adjacent, uninvolved (normal) colonic segments from EOCRC (n=21) and LOCRC (n=22) patients. The EOCRC and LOCRC cohorts were matched for demographic and clinical characteristics. We used The Cancer Genome Atlas Colon Adenocarcinoma (TCGA-COAD) database for validation. We used a series of computational and bioinformatic tools to identify EOCRC-specific differentially expressed genes, molecular pathways, predicted cell populations, differential gene splicing events, and predicted neoantigens. Results: We identified an eight-gene signature in EOCRC comprised of ALDOB, FBXL16, IL1RN, MSLN, RAC3, SLC38A11, WBSCR27 and WNT11, from which we developed a score predictive of overall CRC patient survival. On the entire set of genes identified in normal tissues and tumors, cell type deconvolution analysis predicted a differential abundance of immune and non-immune populations in EOCRC versus LOCRC. Gene set enrichment analysis identified increased expression of splicing machinery in EOCRC. We further found differences in alternative splicing (AS) events, including one within the long non-coding RNA, HOTAIRM1. Additional analysis of AS found seven events specific to EOCRC that encode potential neoantigens. Conclusion: Our transcriptome analyses identified genetic and molecular features specific to EOCRC which may inform future screening, development of prognostic indicators, and novel drug targets.
Overall design	Gene expression profiles of 22 surgically resected tumors and patient-matched adjacent colonic segments from colorectal cancer patients were generated with RNA-sequencing.
Web link	https://doi.org/10.3389/fonc.2024.1365762

GEO accessions: [GSE](#) (series), [GSM](#) (samples), [GPL](#) (platform)

Processed data: Series Matrix / SOFT / tabular files

Raw count data: in supplementary file

Raw reads: usually in **SRA/ENA** as [SRR...](#) → [*.fastq.gz](#)

Raw (FASTQ): larger; needed for alignment/QC

Contributor(s)	Marx OM , Yochum GS , Koltun WA , Mankarious MM			
Citation(s)	Marx OM, Mankarious MM, Koltun WA, Yochum GS. Identification of differentially expressed genes and splicing events in early-onset colorectal cancer. <i>Front Oncol</i> 2024;14:1365762. PMID: 38680862			
NIH grant(s)	Grant ID	Grant title	Affiliation	Name
	R03 CA279861	Wnt/beta-catenin signaling in early-onset colorectal cancer	PENNSYLVANIA STATE UNIV HERSHEY MED CTR	Gregory S. Yochum

Analyze with [GEO2R](#)

Download RNA-seq counts

Submission date	Dec 21, 2023
Last update date	Apr 04, 2025
Contact name	Gregory Yochum
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Department	Biochemistry, Colorectal Surgery
Street address	700 HMC Crescent Road
City	Hershey
State/province	PA
ZIP/Postal code	17033
Country	USA

Platforms (1)	GPL24676 Illumina NovaSeq 6000 (Homo sapiens)
Samples (44)	GSM7988989 24C
More...	GSM7988990 24N
	GSM7988991 27C

Relations	
BioProject	PRJNA1055547

Download family	Format
SOFT formatted family file(s)	SOFT ?
MINiML formatted family file(s)	MINiML ?
Series Matrix File(s)	TXT ?

Supplementary file	Size	Download	File type/resource
GSE251845_RAW.tar	9.0 Mb	(http)(custom)	TAR (of TXT)
GSE251845_htseq_raw_counts.csv.gz	2.1 Mb	(ftp)(http)	CSV

[SRA Run Selector](#) [?](#)

Two ways to download FASTQ

- **#1 `wget` — the universal downloader**
 - Works for **any URL** (FTP, HTTPS)
 - Best when GEO gives you a direct supplementary file link
 - Best when ENA hosts the FASTQ already gzipped (faster than going through SRA)
 - Pros: no install, just works
 - Cons: you need to *know the URL*

Two ways to download FASTQ

- #2 `fastq-dump` (and fasterq-dump) — the SRA-native tool
 - Part of NCBI's **SRA Toolkit**
 - Takes an **SRR accession** directly — no URL needed
 - Knows how to handle paired-end splitting, read filtering, range selection
 - Pros: reproducible, scriptable, handles every SRR uniformly
 - Cons: needs install; first run downloads a cache

Accession

PRJNA1055547



Search

Filters List

- 1 ☐ Bases
- 2 ☐ Bytes
- 3 ☐ create_date
- 4 ☐ source_name
- 5 ☐ tissue
- 6 ☐ tumor_stage

Common Fields

BioProject	PRJNA1055547
Consent	PUBLIC
Assay Type	RNA-Seq
AvgSpotLen	117
Center Name	BIOCHEMISTRY, COLORECTAL SURGERY, PENN STATE COLLEGE OF MEDICINE
Collection_Date	missing
DATASTORE filetype	FASTQ, RUN.ZQ, SRA
DATASTORE provider	GS, NCBI, S3
DATASTORE region	gs.us-east1, ncbi.public, s3.us-east-1

Select

	Runs	Bytes	Bases	Download	Cloud Data Delivery	Computing
Total	44	52.75 Gb	152.51 G	Metadata or Accession List		

☒ ☐	Run	BioSample	age_at_surgery	Bases	Bytes	Experiment	Library Name	create_date	Sample Name	source_name	tissue	tumor_stage
☐ 1	SRR27320655	SAMN39058850	88.75	2.30 G	830.65 Mb	SRX22997924	GSM7989032	2023-12-21 21:33:00Z	GSM7989032	Rectum	Rectum	NA
☐ 2	SRR27320656	SAMN39058851	88.75	3.40 G	1.16 Gb	SRX22997923	GSM7989031	2023-12-21 21:36:00Z	GSM7989031	Rectum	Rectum	1
☐ 3	SRR27320657	SAMN39058852	68	4.74 G	1.63 Gb	SRX22997922	GSM7989030	2023-12-21 21:38:00Z	GSM7989030	Hepatic flexure	Hepatic flexure	NA
☐ 4	SRR27320658	SAMN39058853	68	3.06 G	1.06 Gb	SRX22997921	GSM7989029	2023-12-21 21:34:00Z	GSM7989029	Hepatic flexure	Hepatic flexure	2

Installing SRA Toolkit (optional)

```
# macOS (homebrew)
brew install sratoolkit

# Ubuntu / WSL
sudo apt install sra-toolkit

# Verify
fastq-dump --version
```

The two demo commands

You can find both command in the github repo

- **Download with wget** (from ENA — friendliest mirror):

```
wget ftp://ftp.sra.ebi.ac.uk/vol1/fastq/SRR390/SRR390728/SRR390728.fastq.gz
```

- **Download with fastq-dump** (cap at 10,000 reads for the demo):

```
fastq-dump -X 10000 --split-files --gzip SRR390728
```


GEO help: Mouse over screen elements for information.

Scope: Format: Amount: GEO accession:

Sample GSM7988989[Query DataSets for GSM7988989](#)

Status Public on Apr 11, 2024

Title 24C

Sample type SRA

Source name Sigmoid

Organism [Homo sapiens](#)

Characteristics tissue: Sigmoid
age at_surgery: 52.91666667
tumor stage: 2

Extracted molecule polyA RNA

Extraction protocol Tumors and adjacent colonic segments were surgically resected and aliquots were stored in RNeasy Lysis Buffer (Qiagen) preservative. Samples were homogenized with a pestle and chloroform was added. The samples were centrifuged at 12,000 x g for 15 minutes. The aqueous layer was purified with RNeasy Mini Kit (Qiagen). Samples were selected for poly-adenylation, and cDNA libraries were made and sequenced on MiSeq2500 (Illumina). RNA libraries were prepared for sequencing using standard Illumina protocols

Library strategy RNA-Seq

Library source transcriptomic

Library selection cDNA

Instrument model Illumina NovaSeq 6000

Description 24C

Data processing Base calling with Illumina MiSeq2500
Alignment with STAR version 2.7.3
HTSeq counting of aligned reads
Assembly: hg38
Supplementary files format and content: Count files contain raw counts for sequences aligning to each transcript.

Supplementary file	Size	Download	File type/resource
GSE251845_RAW.tar	9.0 Mb	(http)(custom)	TAR (of TXT)
GSE251845_htseq_raw_counts.csv.gz	2.1 Mb	(ftp)(http)	CSV

[SRA Run Selector](#) 

Raw data are available in SRA

Custom GSE251845_RAW.tar archive:

Supplementary file	File size
☒ GSM7988989_24C_htseq.out.txt.gz	210.5 Kb
☐ GSM7988990_24N_htseq.out.txt.gz	211.3 Kb
☐ GSM7988991_27C_htseq.out.txt.gz	209.0 Kb
☐ GSM7988992_27N_htseq.out.txt.gz	206.9 Kb
☐ GSM7988993_29C_htseq.out.txt.gz	214.4 Kb
☐ GSM7988994_29N_htseq.out.txt.gz	205.9 Kb
☐ GSM7988995_30C_htseq.out.txt.gz	204.9 Kb
☐ GSM7988996_30N_htseq.out.txt.gz	207.2 Kb
☐ GSM7988997_31C_htseq.out.txt.gz	214.6 Kb
☐ GSM7988998_31N_htseq.out.txt.gz	210.4 Kb
☐ GSM7988999_32C_htseq.out.txt.gz	208.7 Kb
☐ GSM7989000_32N_htseq.out.txt.gz	208.3 Kb
☐ GSM7989001_33C_htseq.out.txt.gz	208.7 Kb
☐ GSM7989002_33N_htseq.out.txt.gz	206.4 Kb
☐ GSM7989003_34C_htseq.out.txt.gz	210.6 Kb
☐ GSM7989004_34N_htseq.out.txt.gz	208.2 Kb
☐ GSM7989005_35c_htseq.out.txt.gz	206.3 Kb
☐ GSM7989006_35N_htseq.out.txt.gz	210.2 Kb
☐ GSM7989007_36C_htseq.out.txt.gz	206.5 Kb
☐ GSM7989008_36N_htseq.out.txt.gz	205.8 Kb
☐ Select All	

1 file(s), 210.5 Kb

FTP / HTTPS Download

Pick one tool: `curl -L -O <URL>` or `wget <URL>`

```
ztava@Zahra:~/workshop$ wget https://ftp.ncbi.nlm.nih.gov/geo/series/GSE251nnn/GSE251845/suppl/GSE251845%5Fhtseq%5Fraw%5Fcounts.csv.gz
--2025-10-06 17:26:09-- https://ftp.ncbi.nlm.nih.gov/geo/series/GSE251nnn/GSE251845/suppl/GSE251845%5Fhtseq%5Fraw%5Fcounts.csv.gz
Resolving ftp.ncbi.nlm.nih.gov (ftp.ncbi.nlm.nih.gov)... 130.14.250.12, 130.14.250.7, 130.14.250.31, ...
Connecting to ftp.ncbi.nlm.nih.gov (ftp.ncbi.nlm.nih.gov)|130.14.250.12|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2185236 (2.1M) [application/x-gzip]
Saving to: 'GSE251845_htseq_raw_counts.csv.gz'

GSE251845_htseq_raw_counts.cs 100%[=====>] 2.08M 4.47MB/s in 0.5s

2025-10-06 17:26:10 (4.47 MB/s) - 'GSE251845_htseq_raw_counts.csv.gz' saved [2185236/2185236]
```

```
ztava@Zahra:~/workshop$ curl -L -O https://ftp.ncbi.nlm.nih.gov/geo/series/GSE251nnn/GSE251845/suppl/GSE251845%5Fhtseq%5Fraw%5Fcounts.csv.gz
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           % Done    0      0     793k   0.00:02  0.00:02  --:--:--  793k
```

Exercise 2 — pull real data

You will:

1. Visit GEO page for **GSE251845** and find the supplementary count matrix
2. Download it with `wget`
3. Install or verify `sra-tools`
4. Use `fastq-dump -X 10000` to grab a small SRR
5. Confirm the FASTQ looks right (head, line count)

Answer key: `Exercises/answers/Exercise2_geo_download.sh`

Putting it together on real data

Now that you can download, let's *look* at the data

Two file types, two skill sets:

- **Counts CSV** (gzipped): tabular, comma-separated, has a header
- **FASTQ**: 4 lines per read, no header, sequence-aware

Exercise 3.1 — Counts CSV

Working with `GSE251845_htseq_raw_counts.csv.gz` :

1. Preview the first 10 lines without unzipping → `zcat | head`
2. Count how many genes are in the matrix → `zcat | tail -n +2 | wc -l`
3. Print the header with column numbers → `awk -F','`
4. Extract `gene_id` + two samples → `cut -d',' -f1,3,5`
5. Sort by expression (descending), keep the header on top
6. Count how many genes are zero in your samples

Key idea: `zcat` lets you stream a gzipped file through pipes — no need to unzip first.

Exercise 3.2 — FASTQ inspection

Working with the two paired-end subsets (or your fresh `fastq-dump` output):

1. Count reads → `awk 'NR%4==1' file | wc -l`
 - (Don't use `grep -c '^@'` — quality lines can start with `@` too!)
2. Top 10 starting hexamers (5' end) → `awk 'NR%4==2 { print substr($0,1,6) }' | sort | uniq -c | sort -nr | head`
3. Reads containing ambiguous base N → `awk 'NR%4==2' | grep -c 'N'`
4. Search for a motif (e.g. `ACGTAC`) → `awk 'NR%4==2' | grep -c 'ACGTAC'`
5. Safely grab the first 100 reads → `head -n 400 file > subset_100.fastq`

Key idea: **FASTQ is 4 lines per record** — `NR%4==1` is the header, `NR%4==2` is the sequence, `NR%4==3` is `+`, `NR%4==4` is quality. Once you internalize this, FASTQ stops being scary.

Q&A!!!

With thanks to our collaborators



**Canadian Centre for
Computational
Genomics**

